SEPEHR ABDOUS

sepehrabdous1375@gmail.com

sepehrabdous.github.io

A researcher with a keen interest in **datacenter networking**, developing mechanisms for **high-performance and reliable communication** within and across large-scale datacenters. My work spans both traditional datacenter networks and emerging large-scale AI clusters.

EDUCATION

Ph.D. in Computer Science
Johns Hopkins University
2019–2026 — Baltimore, MD
Thesis: *Taming the Beast: In-network Congestion Management for Large-scale Datacenters*
Advisor: Dr. Soudeh Ghorbani

M.Eng. in Computer Science
Johns Hopkins University
2019–2023 — Baltimore, MD
GPA: 3.81/4.00
Advisor: Dr. Soudeh Ghorbani

B.Sc. in Computer Engineering
Sharif University of Technology
2014–2019 — Tehran, Iran
GPA: 17.06/20.00

SELECTED PUBLICATIONS

Full list here

- Closing the Bandwidth Gap in AI Datacenters with Scalable Multicast
  Scalable multicast, distributed AI training, datacenter networks
  NSDI 2027 • Co-first author
- One to Many: Closing the Bandwidth Gap in AI Datacenters with Scalable Multicast
  AI networking, multicast, collective communication
  HotNets 2025 • First author
- Uno: A One-Stop Solution for Inter- and Intra-Datacenter Congestion Control and Reliable Connectivity
  Congestion control, intra- and cross-datacenter communication, reliable transport
  SC 2025 • Co-first author
- Practical Packet Deflection in Datacenters
  Packet deflection, burst mitigation, programmable networks
  CoNEXT 2023 • First author
- Burst-tolerant Datacenter Networks with Vertigo
  Burst tolerance, packet scheduling, in-network congestion management
  CoNEXT 2021 • Co-first author

SKILLS

Research and Expertise:
Datacenter Networking · AI Infrastructure · Congestion Management · Packet Routing · Network Simulation · Hardware Implementation · AI Networking · Transport Protocols

Coding:
C · C++ · Python · Java

Tools:
OMNeT++ · HTSim · P4 · Git · LaTeX

PROFESSIONAL SERVICES

Reviewer for *Computer Networks*, *Journal of Network and Computer Applications*, *Ad Hoc Networks*, and *IEEE journals* (2022–Present).

EXPERIENCE

Part-time Consulting Researcher
Microsoft
2024–2025 — Baltimore, MD
Designed *Uno*, a unified congestion-control system for reliable communication within and across large-scale datacenter networks.

AI Enablement Intern
CCC Intelligent Solutions
2022 — Baltimore, MD
Developed a high-performance data-migration technique that accelerated transfers between geographically distributed storage systems.

Teaching Assistant
Johns Hopkins University
2019–2025 — Baltimore, MD
Three appointments in *Computer Networks* and *Cloud Computing*.

Research Intern
UCLA REMAP
2018 — LA, CA
Worked with Prof. Jeff Burke's team to integrate Named Data Networking APIs into the Chromium codebase.

Systems Intern
HPDS (High Performance Data Storage)
2018 — Tehran, Iran
Automated system updates and performed system-level storage benchmarking with FIO.

Frontend Developer
Darmaneh
2016–2018 — Tehran, Iran
Developed and maintained Android and iOS applications for an online medical-consultation platform.

Teaching Assistant
Sharif University of Technology
2015–2019 — Tehran, Iran
Fourteen appointments across programming, systems, architecture, numerical methods, and statistics courses.